

Dipartimento di Matematica, Informatica e Geoscienze

CORSO DI LAUREA MAGISTRALE

DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

TESI DI LAUREA MAGISTRALE

TITOLO

Inflation Forecasting with Macroeconomic Indicators: Assessing the Contribution of News Sentiment

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ANNO ACCADEMICO 2025/2026

Abstract

This thesis, developed in collaboration with Generali Insurance Company, assesses whether news-based indicators add predictive information beyond inflation lags and macroeconomic variables for month-on-month inflation in the European Union's fixed 27-country aggregate (EU-27). The project also includes a prototype analytical dashboard intended to support the company's economic analysis.

The European analysis combines Harmonised Index of Consumer Prices (HICP) data, macroeconomic indicators and selected Guardian headlines and short summaries over 2007–2025. FinBERT, a language model adapted to financial sentiment analysis, is used to construct monthly sentiment measures and topic-specific negative-news indicators covering energy, supply chains and geopolitical developments. News published during a given month is combined with lagged inflation and macroeconomic observations to estimate that month's inflation at month-end, ahead of the final HICP release in the following month. This retrospective exercise uses historical data without reconstructing release dates or data vintages.

An autoregressive hidden Markov model (AR-HMM) is compared with linear autoregressive (AR), Random Forest and gradient boosting benchmarks under matched information sets. European model selection uses validation over 2017–2021, with an initial single-block structure comparison and later validation blocks with expanding training samples. The final test covers 2022–2025. A supplementary analysis using U.S. Consumer Price Index (CPI) data and New York Times news provides exploratory context and has no separate validation stage.

In the matched European final test, the AR-HMM combining macroeconomic and news indicators achieves a root mean squared error (RMSE) of 0.3959 percentage points, compared with 0.4460 for its macroeconomic-only counterpart, a reduction of approximately 11.2%. Linear AR performs better during validation, and gains from news vary across specifications and evaluation periods. The findings provide evidence of an incremental predictive contribution from news in the final-test AR-HMM comparison and motivate further development of the dashboard prototype.

Abstract in italiano

Questa tesi, sviluppata in collaborazione con Generali Insurance Company, valuta se gli indicatori ricavati dalle notizie apportino informazione predittiva aggiuntiva rispetto ai valori ritardati dell'inflazione e alle variabili macroeconomiche per l'inflazione su base mensile nell'aggregato a composizione fissa dei 27 Stati membri dell'Unione europea (UE-27). Il progetto comprende inoltre un prototipo di dashboard analitica destinato a supportare l'analisi economica dell'azienda.

L'analisi europea combina dati dell'Indice armonizzato dei prezzi al consumo (HICP), indicatori macroeconomici e titoli e brevi sommari selezionati di The Guardian nel periodo 2007–2025. FinBERT, un modello linguistico adattato all'analisi del sentiment finanziario, viene utilizzato per costruire misure mensili di sentiment e indicatori di notizie negative relativi a energia, catene di approvvigionamento e sviluppi geopolitici. Le notizie pubblicate in un determinato mese vengono combinate con osservazioni ritardate dell'inflazione e delle variabili macroeconomiche per stimare, a fine mese, l'inflazione dello stesso mese, prima della pubblicazione dei dati HICP definitivi nel mese successivo. Questo esercizio retrospettivo utilizza dati storici senza ricostruire le date di pubblicazione o le versioni dei dati disponibili alle diverse date del passato.

Un modello di Markov nascosto autoregressivo (AR-HMM) viene confrontato con modelli autoregressivi lineari (AR), Random Forest e gradient boosting, utilizzando gli stessi insiemi informativi. La selezione dei modelli europei si basa sulla validazione nel periodo 2017–2021, con un confronto iniziale delle strutture su un unico blocco e successivi blocchi di validazione con campioni di addestramento crescenti. Il test finale copre il periodo 2022–2025. Un'analisi integrativa basata sui dati dell'Indice dei prezzi al consumo (CPI) degli Stati Uniti e sulle notizie di The New York Times fornisce un confronto esplorativo e non prevede una fase di validazione separata.

Nel test finale europeo, con modelli valutati sulle stesse date, l'AR-HMM che combina indicatori macroeconomici e notizie raggiunge una radice dell'errore quadratico medio (RMSE) di 0,3959 punti percentuali, rispetto a 0,4460 per la corrispondente specificazione con soli indicatori macroeconomici, con una riduzione di circa l'11,2%. L'AR lineare ottiene risultati migliori in validazione, mentre i miglioramenti dovuti alle notizie variano tra specificazioni e periodi di valutazione. I risultati indicano un contributo predittivo aggiuntivo delle notizie nel confronto tra le specificazioni AR-HMM nel test finale e motivano l'ulteriore sviluppo del prototipo di dashboard.

Acknowledgements

I would like to thank my thesis supervisor for their guidance and feedback throughout this project. Their comments helped me develop the work and think more carefully about the choices behind the analysis.

I am also grateful to my family and friends for their patience and encouragement during my master's studies. I appreciate the support they gave me while I worked on this thesis.

I am particularly grateful to my company supervisor at Generali Insurance Company for the guidance provided in understanding the company’s needs, defining the project requirements and developing a roadmap towards its goals. I also thank Generali Insurance Company for its collaboration and for providing the practical context for investigating inflation estimation with macroeconomic indicators and news sentiment.

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List of Abbreviations

AIC - Akaike information criterion
API - Application programming interface
AR - Autoregressive
AR-HMM - Autoregressive hidden Markov model
ARX - Autoregressive model with exogenous predictors
BERT - Bidirectional Encoder Representations from Transformers
BIC - Bayesian information criterion
CPI - Consumer Price Index
CPU - Central processing unit
CSV - Comma-separated values
ECB - European Central Bank
EU-27 - European Union aggregate of 27 member states
FRED - Federal Reserve Economic Data
GDP - Gross domestic product
HICP - Harmonised Index of Consumer Prices
HMM - Hidden Markov model
HTML - Hypertext Markup Language
MAE - Mean absolute error
ME - Mean error
MS - Markov switching
NLP - Natural language processing
OLS - Ordinary least squares
PPI - Producer Price Index
RMSE - Root mean squared error
RMSFE - Root mean squared forecast error
SA - Seasonally adjusted
SQL - Structured Query Language
WTI - West Texas Intermediate

Chapter 1 Introduction

1.1 Background and Estimation Task

This thesis, developed in collaboration with Generali Insurance Company, assesses whether news-based indicators add predictive information beyond inflation lags and macroeconomic variables for monthly inflation estimation. The project also includes a prototype analytical dashboard for company use, intended to support economic analysis relevant to asset allocation and portfolio management. The principal target is month-on-month inflation in the European Union's fixed 27-country aggregate (EU-27), calculated as the monthly percentage change in the all-items Harmonised Index of Consumer Prices (HICP). The HICP provides a harmonised measure for comparing consumer-price developments across European economies [1].

Recent inflation episodes illustrate the difficulty of predicting price developments when economic conditions change sharply. Chahad et al. (2022) identify unexpected energy-price developments and supply bottlenecks among the factors contributing to Eurosystem and European Central Bank inflation projection errors [2]. These findings motivate examining models that allow inflation dynamics to differ across regimes, while leaving their predictive value to empirical assessment.

The thesis compares an autoregressive hidden Markov model (AR-HMM) with linear autoregressive (AR), Random Forest and gradient boosting benchmarks. The AR-HMM uses a regime-switching framework [3]. In the principal European specification, the intercept and autoregressive coefficients vary across latent states, while the innovation variance is common. Alternative switching restrictions are examined separately. The states represent differences in fitted inflation dynamics; they are not identified structural economic shocks.

The analysis tests the incremental contribution of news indicators alongside macroeconomic predictors. FinBERT, a BERT-based language model adapted to financial sentiment analysis [4], is applied to selected news-text fields to construct monthly sentiment measures. These include general sentiment and topic-specific negative-news indicators for energy, supply chains and geopolitical developments. The European corpus comes from The Guardian, while the supplementary U.S. corpus comes from The New York Times. The indicators describe the tone and topic content of selected news, rather than directly measuring inflationary pressure.

The practical aim is to estimate inflation for a given month at month-end, ahead of the final HICP release in the following month [1]. News indicators summarise articles published during the month being estimated, while inflation and macroeconomic predictors use earlier observations at the experiment-specific lags described in Chapter 3. These lags implement the intended reporting-delay convention. Historical release dates and data vintages are not

reconstructed, so the empirical exercise remains a retrospective month-end nowcast rather than a verified real-time estimate.

Previous studies report mixed evidence on the usefulness of news sentiment for inflation prediction [8, 9]. Their findings concern different countries, inflation measures, horizons and benchmarks. This motivates a setting-specific question: whether Guardian-derived news indicators add predictive information beyond inflation lags and macroeconomic predictors for EU-27 monthly inflation. The thesis addresses this question through comparisons under matched predictor sets and evaluation periods. Accuracy is assessed primarily by root mean squared error (RMSE), which gives greater weight to large estimation errors, and by mean absolute error (MAE), which summarises their average absolute magnitude. Both metrics are expressed in percentage points of inflation.

1.2 Research Questions and Contributions

The main aim is to assess whether news indicators add predictive information beyond inflation lags and macroeconomic variables for EU-27 monthly inflation. Model comparisons and specification choices provide the framework for this assessment. The European analysis addresses three questions:

RQ1. Under matched predictor sets and evaluation periods, does AR-HMM estimate EU-27 month-on-month inflation more accurately than linear AR, Random Forest and gradient boosting?

RQ2. How do regime structure, inflation lag order and calendar-month centring affect AR-HMM estimation accuracy?

RQ3. Do macroeconomic predictors improve accuracy beyond inflation lags, and do target-month news indicators provide further gains beyond inflation lags and macroeconomic predictors?

The project makes three contributions. First, it constructs a monthly dataset combining EU-27 HICP, macroeconomic series and dated news, with sources and processing steps documented to support reproducibility. Second, it compares model structures and predictor groups over matched European evaluation periods to assess the incremental predictive value of macroeconomic and news information. Third, it provides a prototype analytical dashboard for company use. A supplementary U.S. analysis provides exploratory international context.

1.3 Scope and Structure

The principal dataset covers the EU-27 aggregate over 2007–2025. European model selection uses validation over 2017–2021. Initial structure selection evaluates a single validation block; later comparisons use three successive validation blocks with expanding training samples and models fitted once per block. The final test covers January

2022–December 2025. Chapter 4 details this schedule. The supplementary U.S. analysis uses Consumer Price Index (CPI) data and is evaluated over January 2022–September 2025. It has no equivalent separate validation stage, so its findings are exploratory.

The European and U.S. exercises use different inflation measures, news sources and predictor sets. Their error values should therefore be interpreted within each exercise rather than compared directly. Each corpus represents one English-language publication. The use of archived historical series and selected news fields also limits conclusions about operational real-time performance and broader news coverage.

Chapter 2 reviews the related literature. Chapter 3 describes the data and indicator construction, and Chapter 4 specifies the models and evaluation design. Chapter 5 presents and discusses the European results, supplementary U.S. findings, interpretation and limitations. Chapter 6 answers the research questions and identifies future research. Appendix A provides extended results and numerical diagnostics.

Chapter 2 Literature Review

This chapter reviews inflation models and the use of news as an additional information source. It focuses on regime switching, financial sentiment and empirical applications of text-based inflation prediction. These strands motivate the thesis's central question: whether news indicators improve EU-27 monthly inflation estimates beyond inflation lags and macroeconomic predictors.

2.1 Inflation Forecasting and Regime-Switching Models

Chahad et al. (2022) [2] investigated errors in Eurosystem and European Central Bank inflation projections during the pandemic and the subsequent recovery. They compared projections with realised euro-area inflation and examined the contribution of conditioning assumptions, particularly energy prices. Unexpected energy-price increases, supply bottlenecks and reopening effects accounted for much of the underestimation during 2021 and early 2022. Their findings identified energy-price assumptions as a major source of projection error.

Binner et al. (2006) [7] compared linear autoregressive (AR), recurrent neural-network and Markov-switching AR models for quarterly U.S. GDP-deflator inflation over 1960Q1–2003Q1, reserving the final 100 observations for evaluation. Their two-state model switched the intercept and innovation variance while keeping AR coefficients common. At one- and two-quarter horizons, the nonlinear models performed similarly and generally outperformed linear AR; at three and four quarters, Markov switching performed best. These findings motivate testing regime switching, but do not establish its advantage for monthly EU-27 HICP. This thesis therefore evaluates AR-HMM alongside linear AR, Random Forest and gradient boosting under matched predictor sets and evaluation periods.

2.2 Financial Sentiment Analysis and Topic-Based Indicators

Araci (2019) [4] introduced FinBERT, a BERT-based model adapted to financial text through further pretraining and fine-tuning. The study evaluated financial sentiment classification on Financial PhraseBank and scoring on the Financial Question Answering (FiQA) dataset. Ten-fold cross-validation on the full Financial PhraseBank dataset yielded accuracy of 0.86 and an F1 score of 0.84. In this thesis, FinBERT provides sentiment scores for selected news texts; the inflation models then assess their predictive contribution.

Tilly and Livan (2021) [12] combined GDELT news topics and tone with market variables for eight countries. They reduced the news features using partial least squares and evaluated classifiers with walk-forward validation. Their target was the direction of five-day changes in ten-year breakeven inflation rates, predicted one day ahead. Adding news improved F1 scores in seven countries, with Mexico the exception; logistic regression and XGBoost were

strong performers. Although the target differs from realised consumer-price inflation, the study motivates examining topic and tone together.

The news indicators in this thesis cover broader economic developments, including energy, supply chains and geopolitical events. An article need not explicitly mention inflation to contribute to an indicator. Unlike the inflation-pressure classifiers reviewed below, the approach measures news tone and topic-specific negative intensity, then estimates their relationship with inflation empirically. Chapter 3 details their construction.

2.3 News-Based Inflation Forecasting and Nowcasting

Eugster and Uhl (2024) [8] constructed a sentiment index from algorithmically scored news articles and assessed its ability to predict U.S. inflation. Their corpus contained approximately 730,000 articles from 2003Q1 to 2021Q4, and prediction horizons extended to eight quarters. In a pseudo-out-of-sample evaluation, sentiment-based predictions achieved root mean squared errors (RMSEs) approximately 30% below a naive random-walk benchmark, depending on the horizon. The authors also reported stronger performance than macroeconomic and Michigan survey benchmarks at shorter horizons.

Rambaccussing and Kwiatkowski (2020) [9] assessed whether sentiment extracted from economic-policy coverage in U.K. newspapers improved point forecasts of inflation, output and unemployment. They evaluated newspaper-based sentiment measures, including indicators constructed from individual publications and selected keywords. They found no evidence of improved inflation nowcasts or short-term forecasts, although newspaper sentiment was useful for output and unemployment prediction. Performance differed across newspapers and indicator constructions.

Allard et al. (2024) [10] developed an inflation-specific news classifier in a preprint on U.S. inflation nowcasting. They fine-tuned a financial language model to distinguish news indicating rising, falling or neutral inflation pressure and aggregated the classifications into a monthly NEWS index. The index was added to their implementation of a Cleveland Fed-based nowcasting framework. Over January 2020–December 2023, RMSE declined from 0.0409 for the FED baseline to 0.0407 for FED+NEWS. However, the Giacomini–White test returned a p-value of 0.19, so the reduction was not statistically significant at conventional levels.

Boss et al. (2025) [11] investigated euro-area inflation nowcasting with Reddit data in a preprint. They used LLaMA-3-70B-Instruct to classify submissions and comments as indicating upward, downward or neutral inflation pressure, then incorporated the resulting indicators into mixed-frequency regressions. Evaluation covered January 2018–December 2023 against a monthly AR(1) benchmark. The best headline-HICP specification achieved a relative root mean squared forecast error of 0.723, a 27.7% reduction, significant at the 5% level. The best food-inflation specification achieved 0.679, a 32.1% reduction, significant at

the 1% level. These results applied to the best-performing candidate specifications. The evaluation used historical data without reconstructing data vintages.

The findings on text-based inflation prediction are mixed: Eugster and Uhl [8] and Boss et al. [11] reported gains, Allard et al. [10] found a small statistically insignificant reduction, and Rambaccussing and Kwiatkowski [9] found no inflation improvement. Differences in sources, targets, horizons and benchmarks make these gains specific to each setting. The present thesis tests the incremental contribution of news for EU-27 month-on-month HICP by comparing inflation-only, macroeconomic and combined macroeconomic-news predictor sets across four model families. Comparing validation and final-test performance also assesses whether any gains persist across periods. With target-month news and historical data, the principal exercise is retrospective month-end estimation under the alignment rules in Chapter 3.

Chapter 3 Data and Indicator Construction

This chapter explains how the project converts macroeconomic series and news articles into monthly model inputs. It follows four steps: selecting the data sources, defining inflation and macroeconomic predictors, constructing news indicators, and aligning the inputs for evaluation. The main application combines Eurostat data for the European Union's fixed 27-country aggregate with Guardian news. A supplementary U.S. application combines Federal Reserve Economic Data (FRED) series with New York Times news. Both stored datasets span January 2007–December 2025, but transformations, lags and missing observations reduce the samples available to individual models. The two news corpora are processed separately and are not pooled.

3.1 European and U.S. Data Sources and Inflation Measures

The European dataset contains 228 calendar months. All macroeconomic series use Eurostat's EU27_2020 geography: the fixed 27-country composition defined from 2020, applied throughout the historical series. This is an EU aggregate, distinct from the euro area and from EU membership as it changed over time. The project uses Eurostat's published aggregate series rather than an unweighted average of national observations [1, 13].

The principal price series is the all-items Harmonised Index of Consumer Prices (HICP). The source series is not officially seasonally adjusted. The inflation target is its arithmetic month-on-month percentage change:

$$\pi_t = 100\left(\frac{P_t}{P_{t-1}} - 1\right)$$

Here, P□ is the HICP index for reference month t. Monthly inflation is measured in percent per month and is not annualised. Because January 2007 has no preceding price observation in the extracted sample, the dataset contains 227 monthly inflation values, from February 2007 to December 2025.

The U.S. target uses the seasonally adjusted all-items Consumer Price Index for All Urban Consumers (CPIAUCSL), produced by the Bureau of Labor Statistics and retrieved through FRED [6, 14]. Monthly inflation is calculated using the same arithmetic percentage-change formula as for HICP.

The U.S. snapshot also spans 228 calendar months, but October 2025 CPI is missing. This gap prevents calculation of monthly inflation in October and November. December monthly inflation can be calculated, but its first inflation lag is unavailable. The full snapshot therefore contains 225 monthly inflation observations. Although the current experiment's configured endpoint is November 2025, its common evaluation sample contains 45 months, from January 2022 to September 2025.

The U.S. exercise provides supplementary evidence. Differences in price-index coverage, seasonal treatment and news selection limit direct comparisons with the European application.

3.2 Macroeconomic Predictors

The European predictors cover four economic channels: consumer energy prices, producer energy prices, labour-market conditions and industrial activity. Each series has monthly frequency and uses the same EU27_2020 geography as the target [13]. Index growth is measured in percent; changes in rates are measured in percentage points.

Consumer energy prices are measured by the energy component of HICP. Producer energy prices are measured by the domestic energy producer price index (PPI). The latter covers the energy main industrial grouping and is neither seasonally nor calendar adjusted. Both predictors enter as arithmetic monthly percentage changes. They measure prices at different stages of the supply chain and allow the models to test for additional predictive information; their inclusion does not identify a causal transmission effect.

The seasonally adjusted unemployment rate uses Eurostat's total-sex and total-age categories and measures unemployment as a percentage of the labour force. Its first monthly difference enters the models in percentage points.

The industrial production volume index covers mining and quarrying, manufacturing, and electricity, gas, steam and air-conditioning supply. It is seasonally and calendar adjusted and enters as an arithmetic monthly percentage change. These transformations define the model inputs; they do not by themselves establish stationarity or predictive value.

The prepared European dataset stores each transformed macroeconomic predictor at lags of one to six calendar months. A first lag pairs the value for month $t - 1$ with the target for month t. Lags are calculated on the complete monthly calendar before incomplete observations are excluded. Chapter 4 explains which variables and lags each experiment uses. A calendar lag describes temporal alignment; it does not establish whether the observation had been published by the prediction date.

The U.S. predictors are the unemployment rate (UNRATE), effective federal funds rate (FEDFUNDS), industrial production index (INDPRO), and monthly West Texas Intermediate (WTI) spot oil price (MCOILWTICO), all retrieved through FRED [14]. Unemployment and the effective federal funds rate enter as first differences in percentage points. FEDFUNDS is the monthly average of daily effective rates, distinct from the Federal Open Market Committee's target range. Industrial production enters as an arithmetic monthly percentage change.

WTI is measured in U.S. dollars per barrel and enters as 100 times the first difference of its natural logarithm. Non-positive prices are treated as missing before taking logarithms.

Predictor lags depend on the experiment. The U.S. variables represent economic channels related to those in Europe, but the two predictor sets are not identical.

3.3 News Processing and Monthly Sentiment Indicators

The European news corpus is collected through The Guardian Open Platform application programming interface (API) [15]. For each calendar month, the routine searches the Business and World sections together using the fixed query "(eurozone OR Europe) AND (inflation OR oil OR energy OR unemployment OR war OR sanctions OR supply-chain OR shipping)" and retrieves all result pages. After duplicate article identifiers are removed, the saved corpus contains 29,080 articles over 2007–2025. Publication dates determine the monthly assignment. The headline and short summary are cleaned by removing Hypertext Markup Language (HTML) markup, decoding HTML entities and normalising whitespace.

For Guardian articles, FinBERT receives the headline followed by the API's short-summary field, trailText. Full article bodies and keyword tags are excluded from this input; tags are retained only as metadata.

The search query is a reproducible selection rule, not a validated test of inflation relevance or EU-27 coverage. Articles referring to Europe or the eurozone may concern other geographical areas, while relevant country-specific articles may be missed if they lack the query terms. The resulting English-language corpus represents one publication's selected coverage, rather than all news or public sentiment across EU member states.

The supplementary corpus contains 18,063 selected New York Times articles from 2007–2025, retrieved through the Archive API [16]. An article is retained when its combined text and editorial metadata contain at least one predefined macroeconomic term and one predefined U.S.-related term. The searchable fields are the headline, abstract, lead paragraph, snippet, keyword values, news desk and section name. Matching uses case-insensitive substrings. Records with invalid publication dates or empty headlines are excluded, and duplicate identifiers and date–headline pairs are removed.

FinBERT receives the available headline, abstract, lead paragraph and snippet, with identical repeated fragments removed. Full article bodies are excluded. Differences in selection rules and input fields limit direct comparisons of sentiment levels between the New York Times and Guardian corpora.

FinBERT scoring

Bidirectional Encoder Representations from Transformers (BERT) represents words in context by attending to surrounding tokens [18]. For sequence classification, a classifier uses the representation associated with a special classification token to assign a label to the input text. This contextual approach can distinguish expressions that an isolated keyword count would treat alike.

The project uses the existing ProsusAI/finbert checkpoint, associated with Araci's adaptation of BERT to financial text [4, 17]. Its documentation describes further language-model training on a subset of Reuters TRC2 and supervised sentiment training on Financial PhraseBank [19]. This financial focus motivates its use here, but performance on those datasets does not establish classification accuracy for the selected Guardian or New York Times articles.

The checkpoint has 12 encoder layers, 12 attention heads per layer and a hidden dimension of 768 [20]. Its classification layer produces scores for positive, negative and neutral financial sentiment, which softmax converts to three class probabilities.

The project applies the checkpoint in inference mode without additional fine-tuning. Each article's combined text is scored as one sequence, padded as needed and truncated to at most 512 tokens, including special tokens. Text beyond this limit does not affect the classification. The outputs describe financial tone; they are not independently calibrated probabilities of economic events. FinBERT supplies textual indicators, while the subsequent inflation models estimate their predictive relationship with inflation.

Monthly sentiment indicators

For article i, let p_i^+ and p_i^- denote the model's positive and negative class probabilities. The signed sentiment score is:

$$s_i = p_i(\text{positive}) - p_i(\text{negative})$$

The score ranges from −1 to 1. The principal monthly indicator is the unweighted mean of the N_m selected article scores: $S_m = (1/N_m)\Sigma_i s_i$. Each article has equal weight within its month.

The dataset also retains the median score, sample standard deviation, distinct article count, and shares of positive, neutral and negative labels. The predicted label is the class with the highest probability. Thus, the negative-label share is the proportion of articles classified as negative, rather than the average negative probability. The standard deviation measures differences in signed tone across articles, rather than uncertainty about each classification.

The topic-specific indicators combine dictionary matches with FinBERT negative probabilities. Three dictionaries identify geopolitical, energy and supply-chain topics. Example terms are war, invasion and sanctions; oil, natural gas and electricity; and shipping, freight, ports and bottlenecks, respectively. The complete term lists are defined in the project's sentiment-processing module.

Matching uses case-insensitive substrings in the stored sentiment-input text. It uses neither word boundaries nor contextual disambiguation, so both false matches and missed relevant articles are possible. An article may match more than one topic. Dictionary matching uses

the stored text, whereas FinBERT processes at most 512 tokens; the two operations therefore need not use the same text span.

For topic k and reference month t, the negative-intensity indicator is defined as:

$$R_{k,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} I_{i,k}\, p_i(\text{negative})$$

N□ counts all selected articles in month t; p_i^- is article i's negative class probability; and I_i□ is one when the stored text matches topic k and zero otherwise. The denominator is the whole monthly corpus, not only the matching articles.

The indicator ranges from 0 to 1. When a topic has at least one match, it equals the topic's share of articles multiplied by the mean negative probability among those articles. It therefore combines topic prevalence and negative financial tone. Zero means that matching articles make no weighted contribution; it does not denote missing news. Because topic dictionaries overlap, the three indicators cannot be added as mutually exclusive components of total risk.

Financial negativity does not specify the direction of inflation pressure: negative news about weak demand can have different implications from negative news about disrupted energy supply. The topic measures are therefore interpreted as topic-specific negative-intensity indicators. The stored names "Geopolitical_Risk", "Energy_Risk" and "Supply_Chain_Risk" are operational labels, not calibrated risk probabilities or identified economic shocks.

Neither the sentiment classifications nor the topic matches have been validated against a manually annotated inflation-news corpus. Their value in this thesis is assessed through the predictive comparisons described in Chapter 4.

3.4 Data Alignment, Information Availability and Descriptive Statistics

Macroeconomic observations and news indicators are merged by reference month on a regular monthly calendar. Predictor timing depends on the experiment. The initial European forecasting comparisons use prior-month news. The European sentiment-nowcast and final combined comparisons instead use news from target month t together with lagged inflation and macroeconomic predictors. Using target-month news defines a retrospective month-end nowcast, with a later information set than a forecast using only prior-month news.

The supplementary U.S. analysis has the same main objective: to use news published during a month to estimate inflation for that same month. The generic sentiment extensions therefore use same-month news indicators. The project also tests lagged news as alternative inputs: previous-month sentiment in the AR-HMM benchmark, sentiment lags of one to six months in LASSO, and previous-month topic indicators in the subject-specific comparisons.

These additional tests assess whether earlier news also contains useful predictive information.

Missing macroeconomic values are left missing; they are neither replaced with zero nor filled with the previous available value. Monthly changes and lagged predictors are calculated while every calendar month remains in the dataset. For example, a one-month lag for November must refer to October. If the October value is missing, November's lagged predictor is also missing; the September value is not substituted. Observations with a missing target or required predictor are excluded from an experiment's sample only after these calculations.

The European preparation routine requires all five source series and monthly Guardian sentiment to cover the full 228-month calendar; missing coverage raises an error. U.S. news indicators also cover all 228 months, but macroeconomic gaps reduce the usable sample. The number of usable observations depends on the target, transformations and required predictors. Models compared directly are evaluated on common months.

For the European seasonal specifications, the project estimates a separate mean inflation rate for each calendar month from the applicable training period and subtracts that mean from the target. For example, every January observation is centred using the training sample's January mean. This removes the estimated calendar-month mean, including its average inflation component; it is model-specific calendar-month centring rather than an official Eurostat seasonal adjustment.

The same training-derived means are applied to the subsequent evaluation block and added back to predictions before errors are calculated on the original inflation scale. Where predictors are standardised, their means and sample standard deviations are also estimated from training data. These quantities are recomputed for each validation fold and remain fixed within its evaluation block.

European structure selection uses the 2017–2021 validation period. Later comparisons use three validation blocks—2017–2018, 2019–2020 and 2021—with expanding training periods. The final European test contains 48 months, from January 2022 to December 2025. Chapter 4 sets out the model estimation and selection procedures.

Figure 3.1 plots the observed monthly inflation target before model-specific calendar-month centring and shows the pronounced increase during 2022.

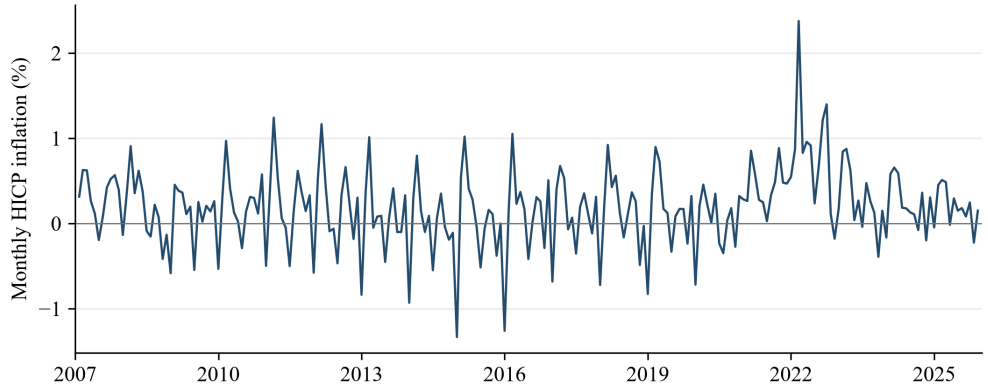


Figure 3.1. EU-27 month-on-month HICP inflation, February 2007–December 2025

Across 228 months, Guardian coverage averages 127.5 articles per month and ranges from 44 to 320. The average monthly mean sentiment is −0.284, and the average monthly negative-label share is 52.4%. The corresponding monthly averages of geopolitical, energy and supply-chain negative intensity are 0.133, 0.037 and 0.066. Each month has equal weight in these summaries; they are not pooled article-level averages.

Differences between the topic indicators reflect dictionary coverage, corpus composition and financial tone. They do not rank causal effects on inflation. Likewise, a negative signed sentiment value does not imply falling consumer prices.

Figure 3.2 shows how the number of selected Guardian articles varies over time.

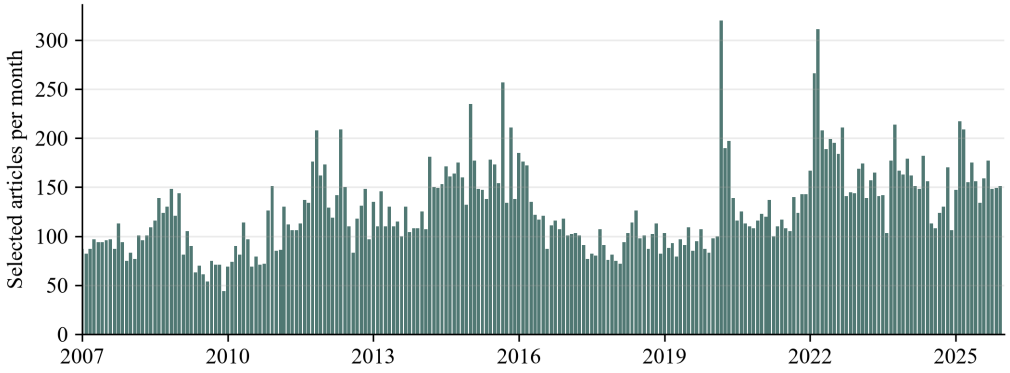


Figure 3.2. Monthly number of selected Guardian articles, January 2007–December 2025

Figure 3.3 shows the monthly mean sentiment and topic-specific negative-intensity indicators, with a separate vertical scale for each panel. The series describes variation within the selected corpus.

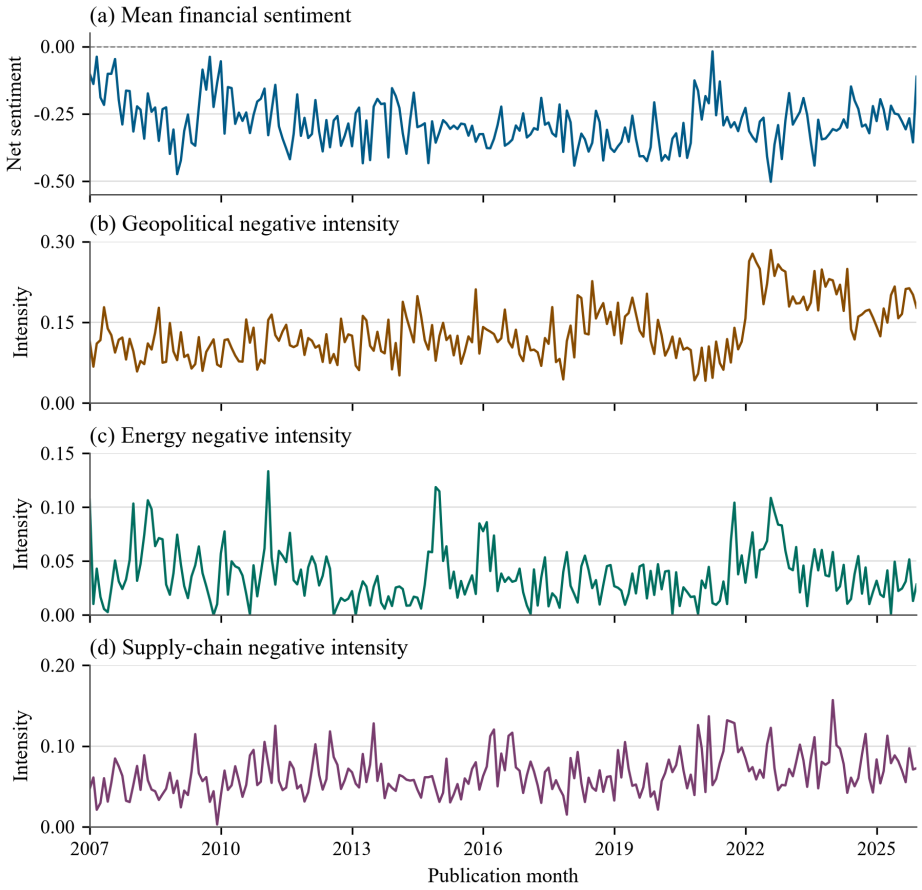


Figure 3.3. Monthly Guardian sentiment and topic-risk indicators, January 2007–December 2025

Chapter 4 Models and Evaluation Design

This chapter explains how the models are used to assess the contribution of news to monthly inflation estimation. The European analysis first selects an AR-HMM structure, then compares inflation-only, macroeconomic and news specifications. Linear AR, Random Forest and gradient boosting provide benchmarks. The U.S. analysis follows the same sequence of predictor comparisons using its available model results.

4.1 Benchmarks: AR, Random Forest and Gradient Boosting

The autoregressive model, AR(p), estimates inflation using its previous p monthly observations. In the equation below, y_t is the inflation target and y_(t−j) is inflation j months earlier. The initial model-selection experiment uses observed monthly inflation; subsequent European comparisons use the calendar-month-centred target defined in Chapter 3.

$$y_t = c + \sum_{j=1}^{p} \phi_j y_{t-j} + \varepsilon_t$$

The intercept c represents the baseline level, φ_j measures the contribution of each inflation lag, and ε_t is the unexplained error. Ordinary least squares estimates the coefficients. AR(1) uses only the previous month. The initial comparison also includes a simple benchmark that repeats the previous month's inflation.

Random Forest and gradient boosting estimate inflation directly from the supplied predictors. They are fitted separately from AR-HMM and do not use its hidden states or state probabilities. The four model families receive the same observed inputs in the matched European comparison.

Random Forest averages predictions from multiple regression trees, each fitted to a resampled training dataset [21]. Gradient boosting builds trees sequentially, with each new tree correcting errors left by the preceding trees [22]. These models provide nonlinear alternatives to linear AR.

For each tree model, three candidate configurations are compared using the validation procedure in Section 4.4. The configuration with the lowest pooled validation RMSE is retained for each predictor group.

4.2 AR-HMM and Alternative Switching Structures

AR-HMM allows the relationship between past and current inflation to change between unobserved states. A Markov transition model describes the probability of remaining in a

state or moving to another state. For the selected two-state model with one inflation lag, the inflation equation is:

$$y_t = c_{S_t} + \varphi_{S_t} y_{t-1} + \varepsilon_t, \qquad S_t \in \{1, 2\}$$

S_t is the state in month t, either 1 or 2. Each state has its own intercept c and lag coefficient φ: state 1 uses c_1 and φ_1, while state 2 uses c_2 and φ_2. The error ε_t has a common variance in the European model [23]. Thus, "two-state AR-HMM(1)" means two possible inflation relationships, each using one previous month.

The initial European experiment compares one, two and three inflation lags, with two or three states for AR-HMM. Candidates are fitted through December 2016 and evaluated over 2017–2021 using observed monthly inflation. Two further variants restrict either the autoregressive coefficient or the mean relationship to be common across states. Table 4.1 reports this comparison.

Table 4.1. European model-structure validation, 2017–2021. Fitting ends in December 2016. RMSE and MAE are in percentage points of monthly inflation.

Specification	RMSE	MAE
AR(1)	0.3895	0.2987
AR-HMM(1), 2 states	0.3775	0.2900
AR-HMM(1), 3 states	0.3844	0.2937
AR(2)	0.3997	0.3030
AR-HMM(2), 2 states	0.3898	0.3054
AR-HMM(2), 3 states	0.3813	0.2855
AR(3)	0.3983	0.3029
AR-HMM(3), 2 states	0.4271	0.3433
AR-HMM(3), 3 states	0.3963	0.3034
AR-HMM(1), 2 states, common AR	0.3862	0.2950
MS-Variance AR(1), 2 states	0.3932	0.2916

The two-state AR-HMM(1) has the lowest validation RMSE among the switching candidates, 0.3775, and is retained for the main European experiments. Selection is based on RMSE; the three-state AR-HMM(2) has a slightly lower MAE. Figure 4.1 shows the state-and-lag comparison.

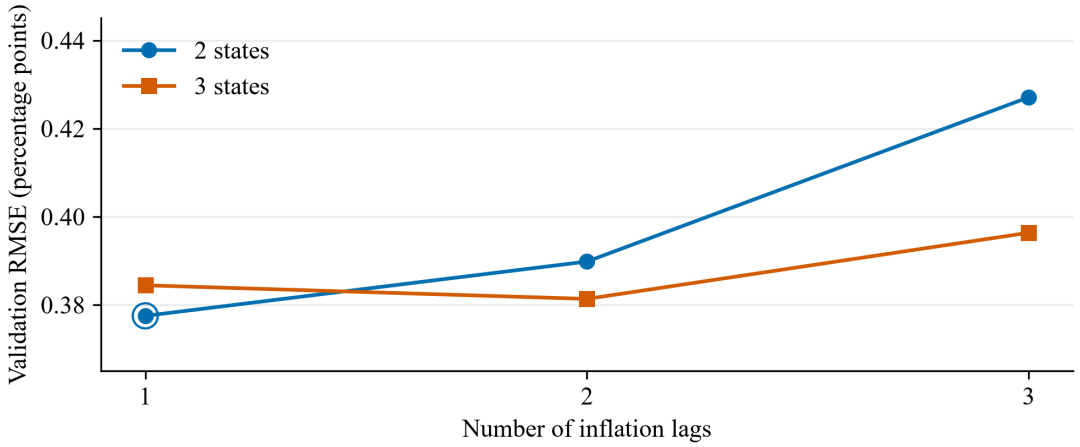


Figure 4.1. Validation RMSE by number of inflation lags and latent states, 2017–2021. The circled point identifies the selected two-state AR-HMM(1).

For each month, the fitted AR-HMM produces an estimate from each state and averages them using predicted state probabilities. These probabilities are calculated before observing that month's inflation. Once inflation is observed, the probabilities are updated for the next month; the fitted coefficients remain fixed within the evaluation period.

Figure 4.2 shows this process for the European model fitted through 2021. The upper panel compares observed inflation with model estimates and marks the most probable state. The lower panel shows both predicted state probabilities. State labels describe the fitted inflation relationships.

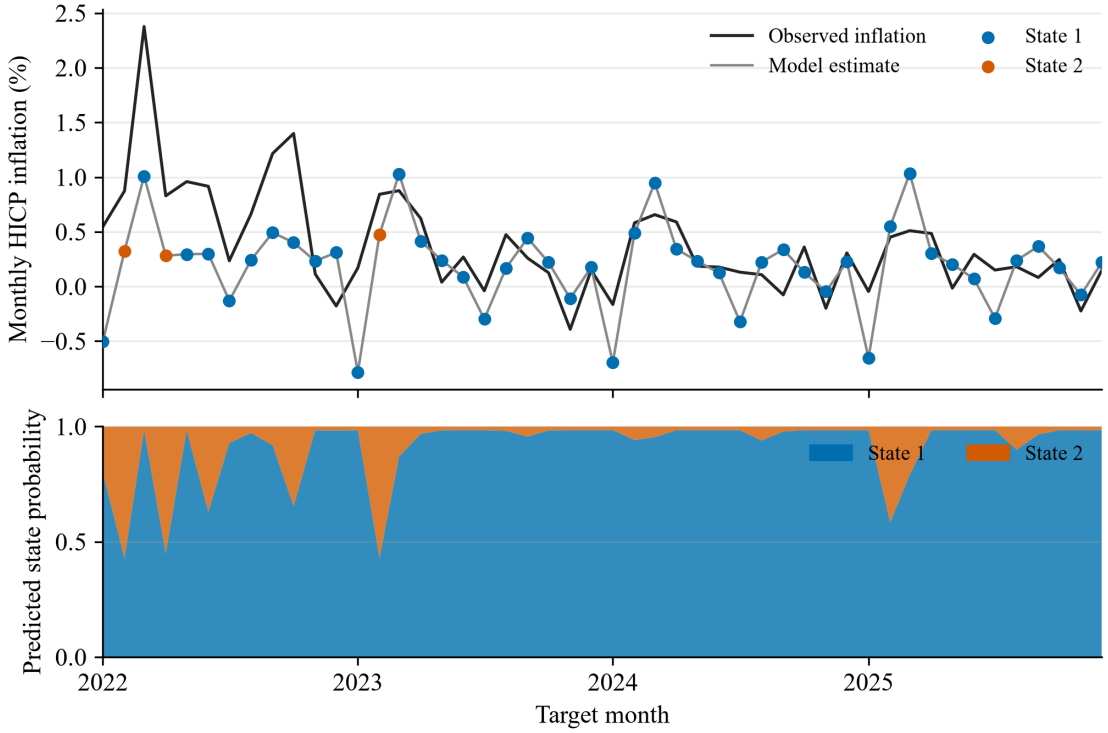


Figure 4.2. European two-state AR-HMM(1), 2022–2025. Top: monthly inflation, model estimates and predicted states. Bottom: state probabilities formed before observing each month's inflation. Calendar effects are restored to the estimates.

4.3 Predictor Groups and Experimental Comparisons

Additional predictors enter the inflation equation through x_t. Their coefficients can differ between AR-HMM states; the linear model uses common coefficients:

$$y_t = c_{S_t} + \sum_{j=1}^{p} \phi_{j,S_t} y_{t-j} + \boldsymbol{\beta}'_{S_t} \boldsymbol{x}_t + \varepsilon_t$$

The vector x_t contains the inputs for target month t. Macroeconomic variables are lagged, while the principal news specifications use articles from the target month. They therefore estimate that month's inflation at month-end.

Before the later European comparisons, calendar-month means are subtracted from inflation using training data only. These effects are added back before errors are calculated. Predictor scaling also uses training data only. Chapter 3 describes these transformations.

The macroeconomic experiments examine unemployment changes, industrial-production growth, energy HICP inflation and energy PPI growth. After an initial lag-1 comparison, lags 1–6 are assessed separately for each indicator using validation RMSE. Table 4.2 gives the selected lags; inflation itself remains at lag 1.

Table 4.2. Selected European macroeconomic predictor lags. Each indicator is assessed at lags 1–6 using combined validation RMSE with the calendar-month-centred two-state AR-HMM(1).

Indicator	Candidate lags	Selected lag (months)
Unemployment	1–6	3
Industrial Production	1–6	1
Energy HICP	1–6	1
Energy PPI	1–6	1

The news experiments compare mean and median sentiment, negative share, dispersion and the three topic-risk indicators. A focused energy comparison tests whether current-month energy-risk news adds information beyond energy PPI growth at lag 1: it compares inflation only, macro only, news only and their combination.

The matched four-family comparison uses the three predictor groups in Table 4.3. It starts with inflation lag 1, adds the four macroeconomic inputs in Table 4.2, then adds current-month mean sentiment and geopolitical, energy and supply-chain indicators. Each group is supplied to every model family.

Table 4.3. Common predictor groups in the European four-family comparison. All groups use the calendar-month-centred inflation target and its first lag.

Group	Inputs supplied for target month t	Model families
Inflation only	Centred inflation, lag 1	Linear AR; AR-HMM; Random Forest; gradient boosting
Inflation + macro	Inflation lag 1; unemployment change lag 3; industrial-production growth, energy HICP inflation and energy PPI growth, each lag 1	Same four families
Inflation + macro + news	All preceding inputs; target-month mean sentiment and geopolitical, energy and supply-chain indicators	Same four families

Comparing families within a group assesses model performance with the same inputs. Comparing predictor groups within a family measures the contribution of macroeconomic information and then news. The macroeconomic lags selected in the AR-HMM experiment are shared across all four families.

4.4 European Validation and U.S. Evaluation Design

Training estimates a model's coefficients or trees. Validation compares candidate settings on later observations. Testing evaluates the chosen specification on a subsequent period. European validation uses 2017–2021, and final testing uses 2022–2025.

Initial state-and-lag selection uses one validation block. Later predictor and tree-model comparisons use three blocks with expanding training samples, as shown in Table 4.4. In each successive block, fitting includes more historical observations.

Table 4.4. European selection and final evaluation. The final specification is chosen using 2017–2021 validation, then fitted through 2021 and tested over 2022–2025.

Step	Fit through	Validation	Final test
State/lag selection	Dec. 2016	2017–2021	—
Validation block 1	Dec. 2016	2017–2018	—
Validation block 2	Dec. 2018	2019–2020	—
Validation block 3	Dec. 2020	2021	—

Selected final model	Dec. 2021	2017–2021 (completed)	2022–2025

The final row uses the settings already chosen during 2017–2021 validation. Once those choices are fixed, the model is refitted using all available observations through December 2021, including the earlier validation years. The 2022–2025 observations are then used for final testing.

Within each evaluation block, coefficients or trees remain fixed while monthly inputs advance. AR-HMM state probabilities update after each observed inflation outcome. The procedure therefore produces successive monthly estimates. Figure 4.3 summarises the fitting and evaluation periods.

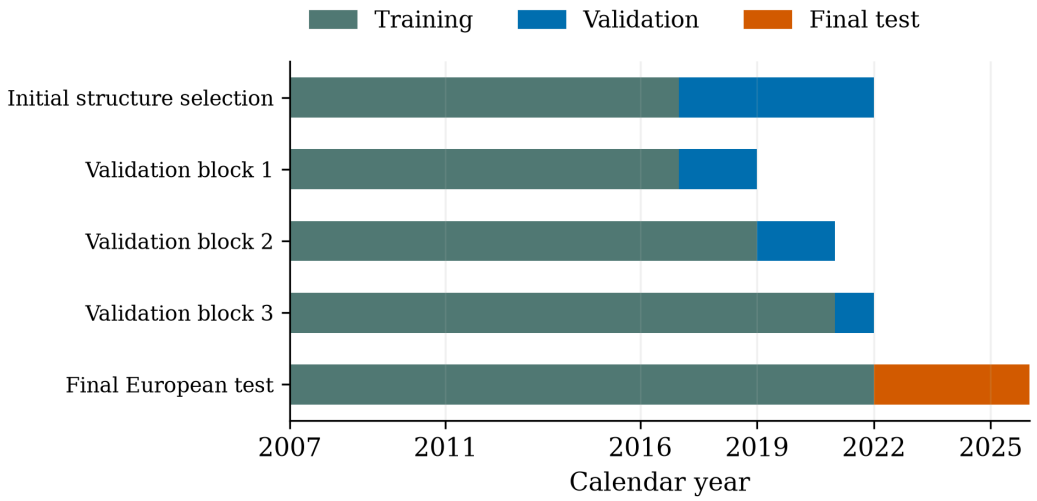


Figure 4.3. European training, validation and final-test periods. Initial structure selection uses one validation block; later comparisons use three expanding-training blocks. The final model is fitted through 2021 after validation has selected its settings.

For each later candidate, the 24, 24 and 12 monthly errors from the three validation blocks are pooled into one RMSE over 60 months. The eligible configuration with the lowest pooled RMSE is retained for each model family and predictor group.

Chapter 5 reports final-test accuracy for 2022–2025. The focused predictor experiments also report 2022 and 2023–2025 separately to show performance during and after the inflation surge.

The U.S. comparison uses a common fitting sample from July 2008 to December 2021 and 45 test months from January 2022 to September 2025. It follows the inflation-only, macroeconomic and news comparison sequence. Because these specifications were compared on the test period without a separate validation stage, the U.S. findings are exploratory.

The U.S. analysis compares linear AR with two-state switching models, examining inflation lags 1–6 and the addition of macroeconomic and news inputs. Generic-news specifications include target-month sentiment, while the separate topic-risk comparison uses lagged indicators. Chapter 5 reports these predictor comparisons alongside the European findings.

Both applications use archived historical data. As discussed in Chapter 3, the principal news results are interpreted as retrospective month-end estimates under the stated input timing.

4.5 Evaluation Metrics and Fair Comparison

Errors are calculated on the original inflation scale. For each scored month, e_t is observed inflation minus the model estimate; T is the number of common scored months. The accuracy measures are:

$$e_t = y_t - \hat{y}_t, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_t^2}$$

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T} |e_t|, \qquad \mathrm{ME} = \frac{1}{T}\sum_{t=1}^{T} e_t$$

RMSE is the main selection measure and gives greater weight to large errors. MAE reports the average absolute error. Lower values indicate better accuracy; both are measured in percentage points. ME measures average bias: positive values indicate underestimation and negative values indicate overestimation.

Relative RMSE gain is 100 × (RMSE_reference − RMSE_model) / RMSE_reference. A positive gain means lower error than the stated reference model. Comparisons use the same target, input timing and scored months.

Chapter 5 uses these measures to assess how adding macroeconomic and news information changes estimation accuracy.

Chapter 5 Results and Discussion

This chapter evaluates whether news improves European monthly inflation estimates beyond inflation lags and macroeconomic indicators. It moves from baseline models and calendar treatment to macroeconomic inputs, news indicators and the matched model comparison. European validation covers 2017–2021 and final testing covers 2022–2025; all errors are in percentage points.

5.1 From Model Development to the Final Test

The main achievement is the full-information AR-HMM's final-test RMSE of 0.3959, the lowest among the 12 matched European specifications. Adding news reduces its error by 11.2% relative to macroeconomic inputs alone. Sections 5.2–5.5 establish the earlier experiments; Section 5.6 presents this result, followed by the supplementary U.S. analysis, interpretation and limitations.

5.2 European Baseline Performance

In the initial uncentred experiment, final-test RMSE falls from 0.5060 for the previous-month benchmark to 0.4984 for linear AR(1), 0.4829 for variance-only switching and 0.4658 for the mean-switching AR-HMM(1). Regime switching improves accuracy within this initial comparison.

The matched calendar-centred benchmark is closer: inflation-only linear AR and two-state AR-HMM both reach RMSE of 0.4331, while Random Forest and gradient boosting give 0.4466 and 0.4523 (Table 5.3). Figure 5.1 compares their estimated paths.

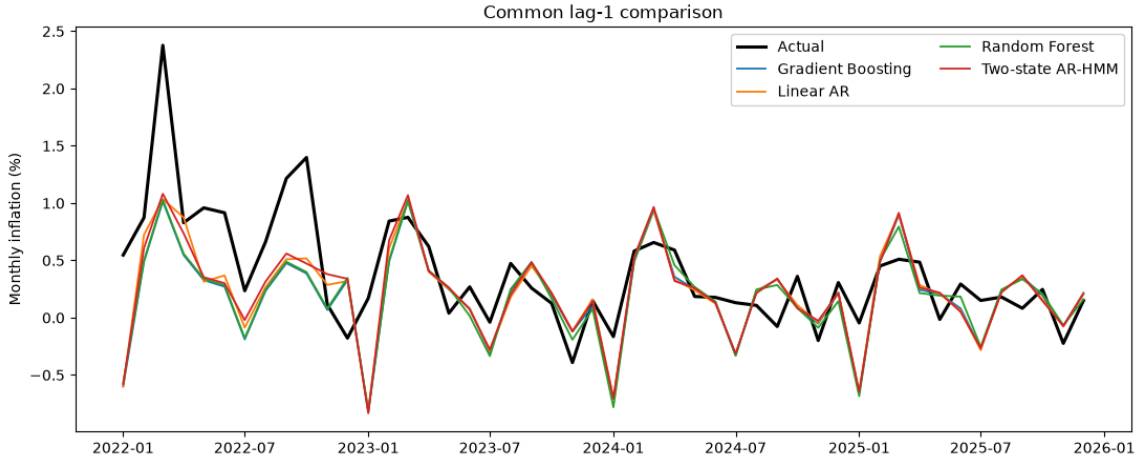


Figure 5.1. EU-27 observed monthly inflation and four model families using the same lag-1 information set, 2022–2025.

All four models underestimate the largest early-2022 increases. This shared difficulty motivates the subsequent tests of calendar treatment and additional information.

5.3 Calendar Treatment and Regime Interpretation

Chapter 4 selects two states and one inflation lag by validation RMSE: 0.3775, compared with 0.3813 for the nearest three-state, two-lag candidate. This two-state AR-HMM(1) is retained for the European predictor experiments.

In the separate seasonal stage, calendar-month centring lowers AR-HMM final-test RMSE from 0.4658 to 0.4553; adding lagged subject risks lowers it to 0.4396. Centred linear AR(1) gives 0.4331. Figure 5.2 illustrates the combined change in calendar treatment and predictors.

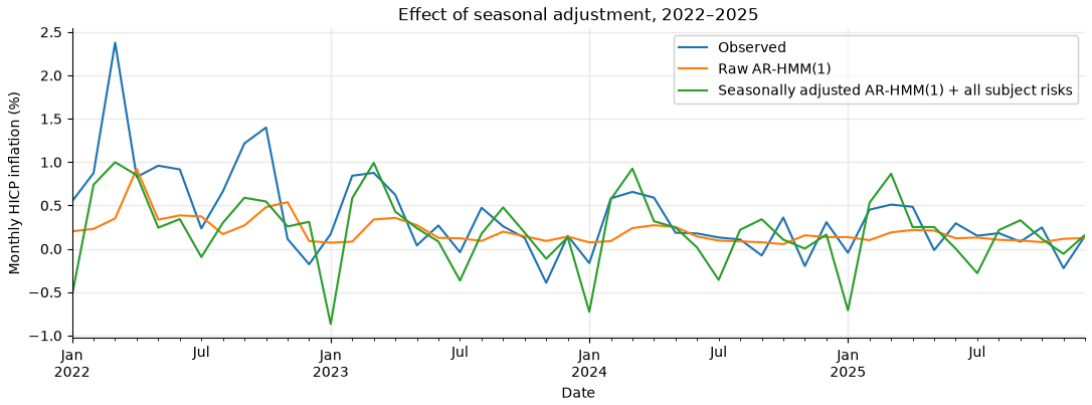


Figure 5.2. Uncentred AR-HMM and the seasonal-stage model with lagged subject risks, 2022–2025.

The European states describe different conditional inflation relationships with a common innovation variance. They should therefore be interpreted as fitted mean states, rather than low- and high-volatility regimes.

5.4 Macroeconomic Predictors

Table 5.1 summarises the macroeconomic additions to the calendar-centred AR-HMM(1). The fixed-lag experiment uses one-month lags; the last two rows report the separate validation-selected-lag experiment. Energy block combines energy HICP and PPI; slow macro combines unemployment and industrial production. Selected-lag order is unemployment, industry, energy HICP and energy PPI.

Table 5.1. European macroeconomic inputs: validation and final-test RMSE, in percentage points. Validation: 2017–2021; test: 2022–2025.

Added inputs	Lag (months)	Validation RMSE	Test RMSE
Baseline	—	0.2416	0.4331
unemployment	1	0.2414	0.4304
industrial production	1	0.2434	0.4342
energy HICP	1	0.2393	0.4674

energy PPI	1	0.2267	0.4594
energy block	1	0.2247	0.4538
slow macro	1	0.2506	0.4356
all macro	1	0.2335	0.4509
Unemployment (selected lag)	3	0.2365	0.4279
All macro (selected lags)	3, 1, 1, 1	0.2444	0.4647

Energy inputs give the strongest validation gains, while unemployment gives the lowest final-test error among the listed additions. Selecting unemployment lag 3 improves RMSE to 0.4279. Figure 5.3 compares the baseline and energy-block estimates during 2022, when both models miss the largest monthly increases.

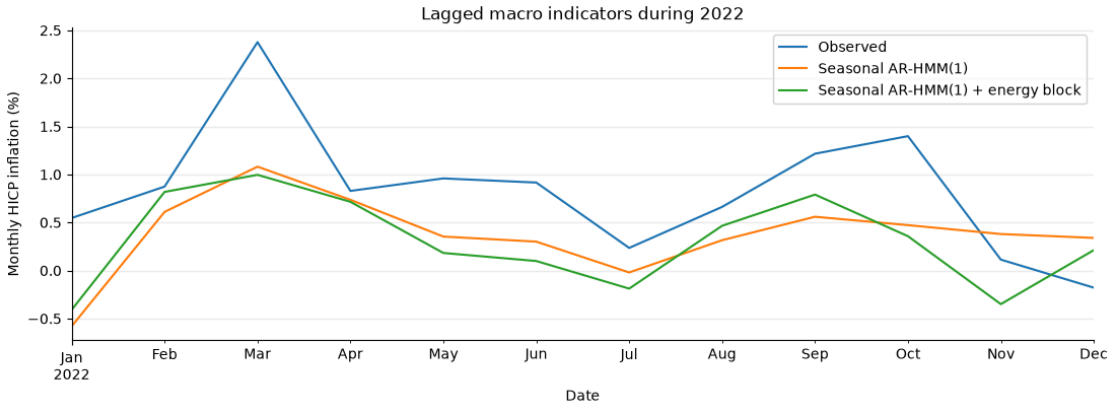


Figure 5.3. European observed inflation and AR-HMM estimates with and without the lagged energy block during 2022

5.5 News Sentiment and Subject-Specific Risk

Table 5.2 compares target-month Guardian indicators added to the inflation-lag baseline. These are month-end estimates using news published during the month being estimated.

Table 5.2. European news additions: validation and final-test RMSE, in percentage points. Calendar-centred AR-HMM(1); validation: 2017–2021; test: 2022–2025.

Added news indicator	Validation RMSE	Test RMSE
Baseline	0.2408	0.4331
mean sentiment	0.2405	0.4374
median sentiment	0.2434	0.4347
negative share	0.2462	0.4335
sentiment dispersion	0.2417	0.4383
geopolitical risk	0.2485	0.4477

energy risk	0.2395	0.4337
supply-chain risk	0.2407	0.4263
all subject risks	0.2557	0.4156

Generic sentiment produces little change in final-test RMSE. The joint topic-risk model has the lowest test error in this experiment, at 0.4156, approximately 4.0% below baseline; energy risk has the lowest validation error.

The focused energy-PPI and energy-risk experiment gives test RMSE of 0.4331 for baseline, 0.4594 for macro only, 0.4337 for news only and 0.4743 for both (Figure 5.4; Table A.4). This narrower combination does not improve accuracy.

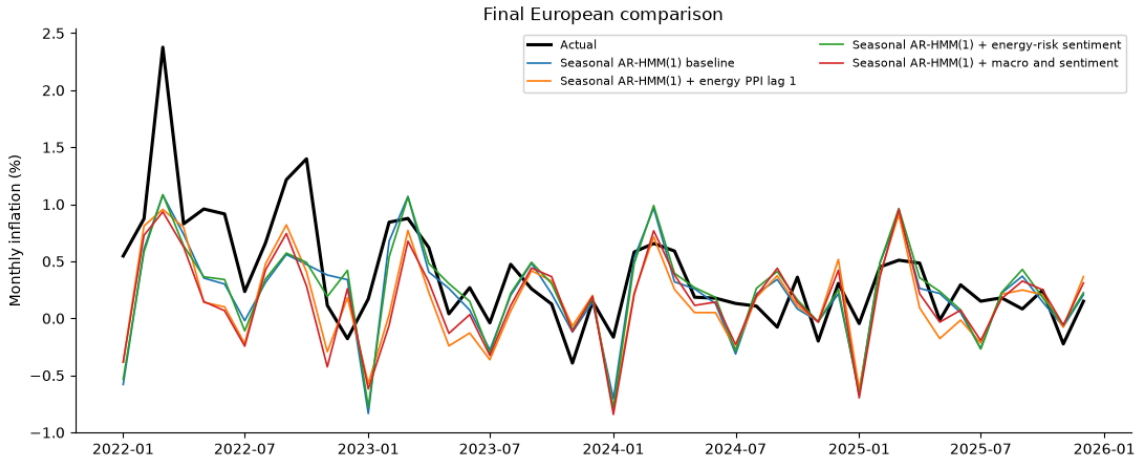


Figure 5.4. Focused European baseline, energy-PPI, energy-risk and combined estimates, 2022–2025.

5.6 Matched Comparison and the Main Final-Test Result

Table 5.3 compares four independent model families using the same inflation-only, macroeconomic, and macroeconomic-plus-news information sets. Within each group, the predictors, calendar treatment and scored months are shared. Figure 5.5 summarises the validation and final-test rankings.

Table 5.3. Matched European RMSE by model family and information set, in percentage points. Validation: 2017–2021 (60 months); final test: 2022–2025 (48 months), also divided into 2022 and 2023–2025.

Inputs / model	Validation	2022–2025	2022	2023–2025
Inflation / Linear AR	0.2414	0.4331	0.6872	0.3044
Inflation / AR-HMM	0.2416	0.4331	0.6820	0.3083
Inflation / Random Forest	0.2515	0.4466	0.7193	0.3057
Inflation / Gradient Boosting	0.2441	0.4523	0.7302	0.3084

+ macro / Linear AR	0.2357	0.4585	0.7051	0.3385
+ macro / AR-HMM	0.2742	0.4460	0.6686	0.3409
+ macro / Random Forest	0.2533	0.4622	0.7323	0.3257
+ macro / Gradient Boosting	0.2470	0.4637	0.7385	0.3238
+ macro + news / Linear AR	0.2356	0.4556	0.7299	0.3149
+ macro + news / AR-HMM	0.2432	0.3959	0.5960	0.3009
+ macro + news / Random Forest	0.2476	0.4615	0.7594	0.3029
+ macro + news / Gradient Boosting	0.2446	0.4665	0.7831	0.2928

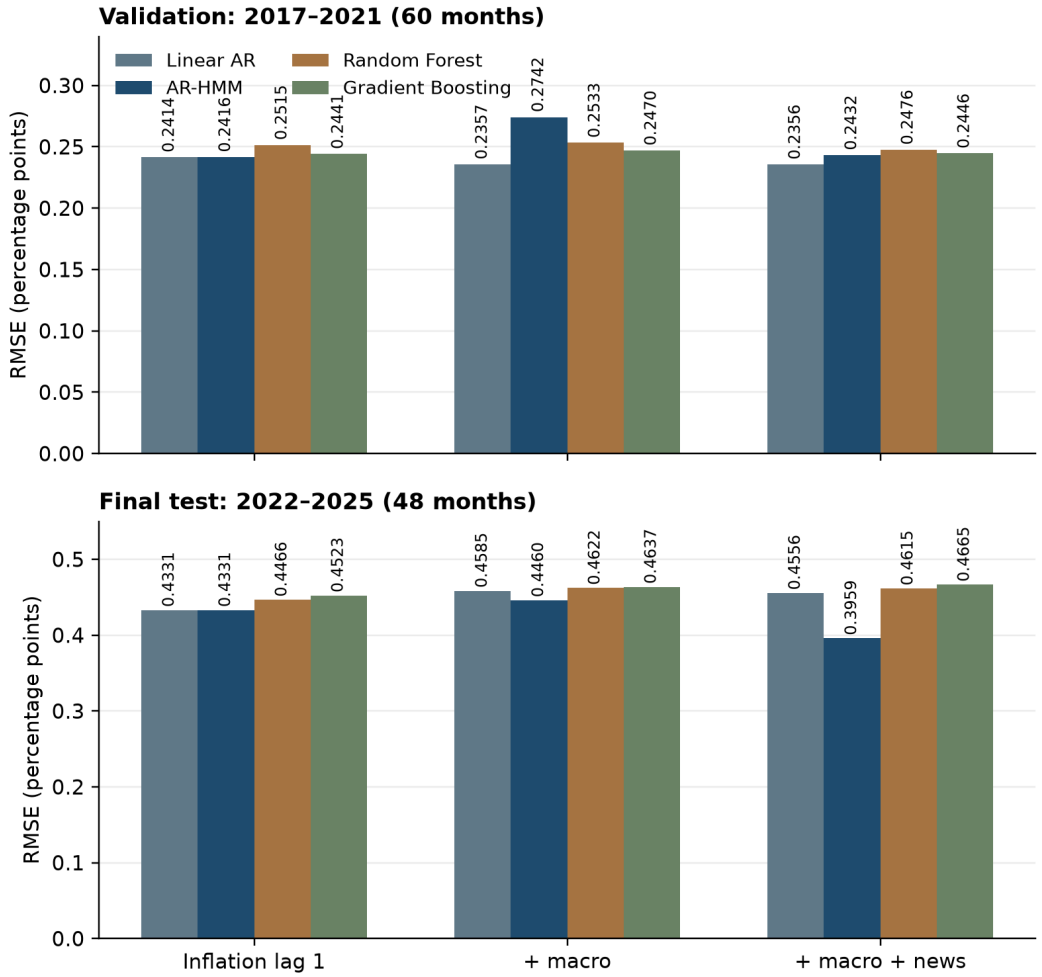


Figure 5.5. Matched European validation and final-test RMSE by information set. Validation: 2017–2021 (60 months); final test: 2022–2025 (48 months). Errors are in percentage points. The full-information AR-HMM has the lowest final-test RMSE, at 0.3959.

The full-information AR-HMM achieves the lowest final-test RMSE among all 12 matched specifications, at 0.3959, with MAE of 0.3059. Adding news to its macroeconomic specification reduces RMSE from 0.4460 by approximately 11.2%; the gain relative to its inflation-only model is approximately 8.6%. These reductions use archived rounded scores.

The improvement occurs in both 2022 and 2023–2025, with the larger gain in 2022 (Table 5.3). Linear AR leads validation in each information set, while the enriched AR-HMM leads the full final test.

5.7 Supplementary U.S. Results

The supplementary U.S. analysis follows the same reporting sequence: inflation baselines, macroeconomic inputs and news additions. Table 5.4 summarises selected results over January 2022–September 2025 (45 months). Unlike Europe, this exercise has no separate validation stage.

Table 5.4. Supplementary U.S. monthly-inflation results, in percentage points. Static fits through 2021. AR-HMM rows distinguish unrestricted switching from variance-only switching.

Specification	Test RMSE	Test MAE
Linear AR(1)	0.263740	0.182455
Unrestricted AR-HMM(1)	0.303487	0.197187
Variance-only AR-HMM(1)	0.261231	0.180262
Variance-only + unemployment + industry	0.261180	0.180250
Linear AR + mean sentiment	0.263938	0.183034
Linear AR + sentiment dispersion	0.261011	0.180731
Variance-only + mean sentiment	0.261701	0.181416
Variance-only + all lagged topic risks	0.274330	0.185814

The variance-only AR-HMM improves on unrestricted switching and slightly outperforms linear AR. Adding unemployment and industry changes its RMSE only marginally. News additions do not improve the variance-only baseline; linear AR benefits modestly from sentiment dispersion.

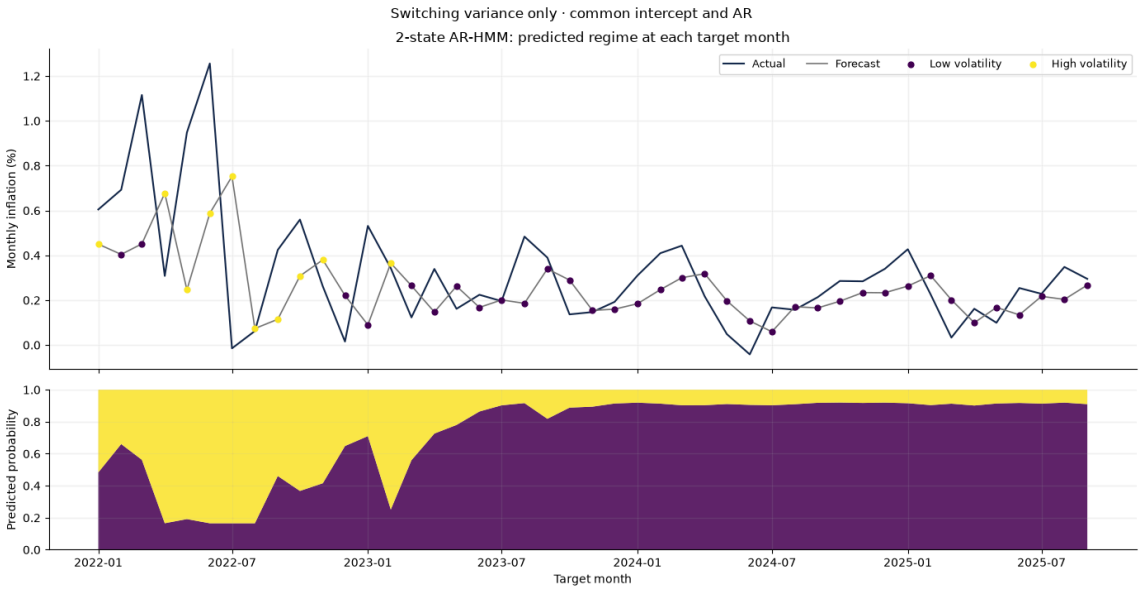


Figure 5.6. U.S. variance-only model estimates, predicted states and probabilities

Figure 5.6 shows higher predicted volatility during much of 2022 and lower volatility later. These variance states differ from the European conditional-mean states.

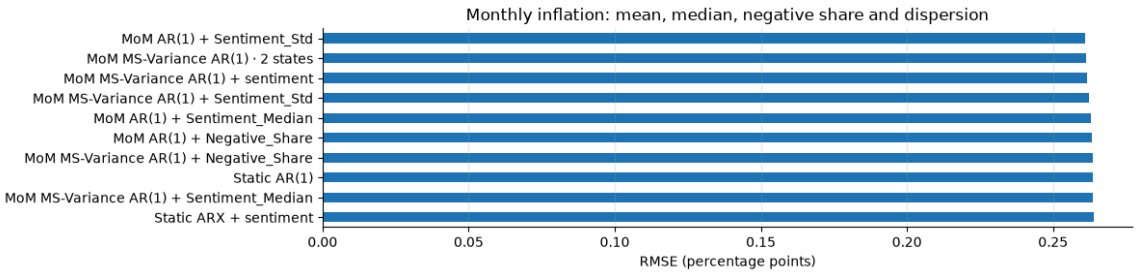


Figure 5.7. U.S. monthly inflation with alternative sentiment aggregations

Figure 5.7 compares sentiment aggregations. The European full-information news gain is the main finding; the U.S. results provide supplementary context under a different target, news source and evaluation design.

5.8 What the Results Establish

The European analysis demonstrates an incremental predictive contribution from news in the matched AR-HMM: adding the news block to macroeconomic inputs lowers final-test RMSE by approximately 11.2% and produces the lowest error among the 12 matched specifications. The gain is visible in both reported subperiods. This directly answers the thesis's central question about the information added by news.

The tables also show that the benefit emerges from the broader information set: individual sentiment measures and the focused energy combination do not deliver the same improvement. For the company's analytical dashboard, the full-information AR-HMM is therefore a useful specification for further assessment alongside the linear AR benchmark.

5.9 Study Limitations

The main practical constraint was access to suitable historical news. Bloomberg data and API access were unavailable during the thesis, so the analysis used Guardian and New York Times articles. The company plans to provide further news resources for future work. The present results measure the information in the collected articles; broader country, sector and event coverage remains to be assessed.

The evaluation used historical economic values as downloaded, rather than a record of the releases available at each past month-end. It therefore assesses historical predictive accuracy. Testing the dashboard with time-stamped incoming news and economic releases is the next step toward measuring live performance. The U.S. exercise remains supplementary because its data and evaluation design differ from Europe's.

Chapter 6 Conclusion

6.1 Answers to the Research Questions and Contributions

This thesis assessed whether regime switching, macroeconomic predictors and news indicators improve EU-27 month-on-month inflation estimation. The principal news comparisons are retrospective month-end nowcasts, with target-month articles combined with lagged inflation and macroeconomic inputs.

For RQ1, AR-HMM improves accuracy relative to the other model families in the matched final test when it receives the full macroeconomic and news information set. Its 2022–2025 RMSE is 0.3959 percentage points, compared with 0.4556 for linear AR, 0.4615 for Random Forest and 0.4665 for gradient boosting. It has the lowest RMSE among all 12 matched specifications and MAE of 0.3059.

For RQ2, the initial European validation selects two states and one inflation lag under the stated RMSE criterion. More states or longer lag orders do not consistently improve accuracy. Calendar-month centring is useful in the archived seasonal experiment: final-test AR-HMM RMSE falls from 0.4658 without centring to 0.4553 with centring. That experiment retains its own baseline because preprocessing and separately estimated fits differ from the later matched comparison; its scores should not be combined with the matched 0.3959 result to attribute a single cumulative gain.

For RQ3, macroeconomic predictors alone do not consistently improve accuracy beyond inflation lags, but adding the news block provides a clear observed gain within the matched AR-HMM. Its final-test RMSE rises from 0.4331 with inflation alone to 0.4460 with macroeconomic inputs, then falls to 0.3959 when target-month mean sentiment and the three topic-risk indicators are added. The news addition reduces RMSE by 0.0501 percentage points, approximately 11.2%, relative to the macro-only specification. The full-information model is also approximately 8.6% below the inflation-only AR-HMM.

This answer remains specific to the combined information set. The narrower experiments give mixed results: energy variables improve validation but worsen final-test accuracy, unemployment offers a modest gain, generic sentiment does not consistently reduce full-test RMSE, and the focused energy-PPI and energy-risk combination performs worse than its baseline (Sections 5.4–5.5). The full-information AR-HMM improves on its inflation-only version in both 2022 and 2023–2025, with a larger gain in 2022.

The thesis contributes a documented monthly dataset linking EU-27 inflation, macroeconomic series and dated Guardian news; an evaluation framework for comparing model structures and predictor groups on matched dates; and a prototype analytical dashboard for company use. The supplementary U.S. analysis provides exploratory context, with limited and specification-dependent gains from sentiment additions. The main

European finding is that the broader news block improves the matched macroeconomic AR-HMM over the final-test period. This supports retaining that specification for further assessment alongside simple AR benchmarks, with conclusions bounded by the available news sources and retrospective design.

6.2 Future Research

The first priority is to use the news resources that the company plans to provide. Their suitability should be checked for historical coverage, usable article content, publication timestamps and consistent access. Once suitable resources are available, they can be connected to the existing preparation and sentiment-scoring workflow. This directly addresses the principal data-access constraint encountered during the thesis.

The next step is to test whether the company-provided sources add predictive value. The initial comparison should retain the current sentiment and topic-risk definitions, model specifications and evaluation periods so that the effect of changing the news source can be assessed clearly. Linear AR and the current full-information AR-HMM should remain reference models. More detailed country or sector filters should be introduced only where the supplied metadata and article coverage support them; broader access should not be assumed to improve accuracy.

A practical operational extension is to connect the supplied news feed to the prototype dashboard and generate estimates at a fixed monthly cut-off. Incoming articles and economic releases should be checked for missing or duplicate records, and each estimate should retain a snapshot of the inputs available at that time. Subsequent comparison with released inflation would then assess performance under actual information availability and show whether the retrospective gains persist in company use.

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Appendix A Extended Results and Diagnostics

A.1 Reading the extended results

European validation covers 2017–2021 (60 months) and testing 2022–2025 (48 months), using calendar-centred inflation. The exploratory U.S. test covers January 2022–September 2025 (45 months), without independent validation. U.S. models fit through 2021 unless labelled expanding.

RMSE, MAE and ME are in percentage points; ME = actual − estimate. Table A.1 reports scores to four decimal places; other tables retain saved precision. Experiments retain separate baselines.

A.2 European extended results

Table A.1. Matched European model families: validation and final-test accuracy

Inputs / family	Val. RMSE	Val. MAE	Test RMSE	Test MAE
I / Linear AR	0.2414	0.1613	0.4331	0.3230
I / AR-HMM	0.2416	0.1610	0.4331	0.3263
I / Random Forest	0.2515	0.1781	0.4466	0.3322
I / Gradient Boosting	0.2441	0.1658	0.4523	0.3400
I+M / Linear AR	0.2357	0.1569	0.4585	0.3529
I+M / AR-HMM	0.2742	0.1930	0.4460	0.3479
I+M / Random Forest	0.2533	0.1747	0.4622	0.3503
I+M / Gradient Boosting	0.2470	0.1677	0.4637	0.3541
I+M+N / Linear AR	0.2356	0.1532	0.4556	0.3402
I+M+N / AR-HMM	0.2432	0.1727	0.3959	0.3059
I+M+N / Random Forest	0.2476	0.1655	0.4615	0.3467
I+M+N / Gradient Boosting	0.2446	0.1637	0.4665	0.3440

I: inflation lag 1; M: four macroeconomic inputs at the lags in Table 4.2; N: target-month mean sentiment and three topic risks. MAE extends Table 5.3. Matched monthly predictions and ME were not saved.

Table A.2a. European fixed-lag macroeconomic inputs

Added inputs	Val. RMSE	Test RMSE	Test MAE	Test ME
Baseline	0.241558	0.433122	0.326277	0.156324
+ unemployment	0.241450	0.430356	0.320507	0.160460
+ industrial production	0.243411	0.434226	0.326379	0.156287
+ energy HICP	0.239292	0.467408	0.339938	0.206688
+ energy PPI	0.226721	0.459437	0.356988	0.230968
+ energy block	0.224659	0.453766	0.351012	0.224491
+ slow macro	0.250581	0.435583	0.325532	0.162293
+ all macro	0.233467	0.450851	0.352108	0.223627

All macroeconomic lags are 1. Energy block: energy HICP and PPI; slow macro: unemployment and industrial production. RMSE summary: Table 5.1.

Table A.2b. European target-month news indicators

Added inputs	Val. RMSE	Test RMSE	Test MAE	Test ME
Baseline	0.240849	0.433122	0.326277	0.156324
+ mean sentiment	0.240487	0.437417	0.331426	0.159011
+ median sentiment	0.243358	0.434684	0.329155	0.154899
+ negative share	0.246152	0.433505	0.330174	0.152270
+ sentiment dispersion	0.241725	0.438252	0.324025	0.167199
+ geopolitical risk	0.248537	0.447707	0.333311	0.187185
+ energy risk	0.239496	0.433661	0.328784	0.142512
+ supply-chain risk	0.240711	0.426314	0.313789	0.149190
+ all subject risks	0.255729	0.415643	0.307939	0.122422

RMSE summary: Table 5.2.

Table A.3. European validation-selected macro lags

Specification	Validation RMSE	Test RMSE	Test MAE	Test ME
Baseline	—	0.433122	0.326277	0.156324
+ unemployment lag 3	0.236519	0.427897	0.318896	0.150488
+ industrial production lag 1	0.242578	0.442905	0.344964	0.156083
+ energy HICP lag 1	0.234209	0.467399	0.339898	0.206727
+ energy PPI lag 1	0.226721	0.459437	0.356989	0.230968
+ selected macro lags	0.244375	0.464662	0.359972	0.232087

Lags are selected from 1–6; the joint model combines all four. Baseline validation RMSE was not saved (—). This experiment is separate from Table A.1.

Table A.4. Focused European energy-PPI and energy-risk comparison

Specification	Validation RMSE	Test RMSE	Test MAE	Test ME
Baseline	0.240849	0.433122	0.326277	0.156324
+ energy PPI lag 1	0.226721	0.459438	0.356989	0.230969
+ energy-risk sentiment	0.239496	0.433662	0.328784	0.142514
+ macro and sentiment	0.266516	0.474285	0.358637	0.225282

Energy PPI uses lag 1; energy risk uses the target month. This focused design differs from the matched nine-predictor comparison (Section 5.6).

A.3 Supplementary U.S. results

Table A.5. U.S. baseline, expanding estimation and switching restrictions

Specification	RMSE	MAE	ME
Static AR(1)	0.263740	0.182455	0.067387
Static ARX + sentiment	0.263938	0.183034	0.069157
Static HMM	0.302536	0.197173	0.142413
Static AR-HMM	0.303487	0.197187	0.053888
Static AR-HMM + lagged sentiment	0.261486	0.181474	0.070805
Static LASSO	0.267084	0.185420	0.083598
Static LASSO + sentiment	0.267930	0.183599	0.099399
Expanding AR(1)	0.265544	0.181674	0.054634
Expanding AR-HMM(1)	0.267127	0.185508	0.063486
Switching intercept · common AR · common variance	0.261659	0.180324	0.074924
Switching intercept + variance · common AR	0.264829	0.181883	0.072158
Switching variance only · common intercept and AR	0.261231	0.180262	0.063102

Expanding models refit monthly. HMM has two state-specific means, common variance and no inflation lags. LASSO (least absolute shrinkage and selection operator) uses inflation lags 1–6, adding sentiment lags 1–6 in its extension; the penalty is selected by chronological cross-validation within training data. ARX uses target-month sentiment; the AR-HMM sentiment extension uses lag 1. Selected results appear in Table 5.4.

Table A.6. U.S. monthly lag-order comparison

Specification	RMSE	MAE	ME
Static AR(1)	0.263740	0.182455	0.067387
MS-Variance AR(1)	0.261231	0.180262	0.063102
AR(2)	0.276363	0.192375	0.085438
MS-Variance AR(2)	0.271161	0.186523	0.089523
AR(3)	0.276341	0.192362	0.085417
MS-Variance AR(3)	0.277484	0.189833	0.095695
AR(4)	0.276972	0.192142	0.079834
MS-Variance AR(4)	0.270144	0.184350	0.071245
AR(5)	0.283592	0.195195	0.089243
MS-Variance AR(5)	0.268158	0.183366	0.065695
AR(6)	0.284635	0.195592	0.092447
MS-Variance AR(6)	0.280838	0.190575	0.094274
naive	0.304674	0.200200	-0.008787

MS-Variance has common intercept and slopes with state-specific innovation variance. The naïve estimate uses the previous month's inflation.

Table A.7. U.S. monthly inflation: all 16 macroeconomic information sets

Added macro inputs	AR RMSE	MS-var. RMSE	AR MAE	MS-var. MAE
Inflation lag 1 only	0.263740	0.261231	0.182455	0.180262
Oil	0.277795	0.277911	0.202608	0.200310
Unemployment	0.263928	0.261268	0.182532	0.180310
Industry	0.263753	0.261216	0.182372	0.180250
Fed rate	0.286857	0.261561	0.208379	0.183880
Oil + Unemployment	0.282465	0.282597	0.207562	0.206024
Oil + Industry	0.280415	0.277069	0.205540	0.199110
Oil + Fed rate	0.275599	0.277015	0.204555	0.200238
Unemployment + Industry	0.264054	0.261180	0.182832	0.180250
Unemployment + Fed rate	0.297396	0.261549	0.217248	0.183856
Industry + Fed rate	0.296181	0.261571	0.215252	0.183982
Oil + Unemployment + Industry	0.282325	0.282485	0.207525	0.205954
Oil + Unemployment + Fed rate	0.284570	0.278313	0.212759	0.204343
Oil + Industry + Fed rate	0.283106	0.275551	0.209407	0.200097
Unemployment + Industry + Fed rate	0.298173	0.261489	0.217574	0.183886
Oil + Unemployment + Industry + Fed rate	0.285189	0.278013	0.212340	0.204161

All subsets include inflation lag 1; macroeconomic inputs also use lag 1. MS-var. denotes the variance-only structure defined below Table A.6.

Table A.8. U.S. monthly sentiment extensions

Specification	RMSE	MAE	ME
Static AR(1)	0.263740	0.182455	0.067387
Static ARX + sentiment	0.263938	0.183034	0.069157
AR(1) + Industry	0.263753	0.182372	0.066789
AR(1) + Industry + sentiment	0.263969	0.183095	0.068537
MS-Variance AR(1)	0.261231	0.180262	0.063102
MS-Variance AR(1) + sentiment	0.261701	0.181416	0.066483
MS-Variance AR(1) + Unemployment + Industry	0.261180	0.180250	0.063726
MS-Variance AR(1) + Unemployment + Industry + sentiment	0.261672	0.181449	0.067465
AR(1) + median sentiment	0.262983	0.181389	0.063065
AR(1) + negative share	0.263294	0.181940	0.065019
AR(1) + sentiment dispersion	0.261011	0.180731	0.059421
MS-Variance AR(1) + median sentiment	0.263747	0.183754	0.074704
MS-Variance AR(1) + negative share	0.263567	0.182946	0.075192
MS-Variance AR(1) + sentiment dispersion	0.262231	0.180881	0.066905

News measures use the target month; industry and unemployment use lag 1. Dispersion is the sample standard deviation of article-level net sentiment.

Table A.9. U.S. lagged subject-risk experiment (separate saved bundle)

Specification	RMSE	MAE	ME
Static AR(1)	0.263740	0.182455	0.067387
AR(1) + geopolitical risk	0.262466	0.181354	0.064620
AR(1) + energy risk	0.277336	0.186378	0.090952
AR(1) + supply-chain risk	0.265324	0.183463	0.069372
AR(1) + all subject risks	0.275076	0.184033	0.086929
MS-Variance AR(1)	0.261231	0.180262	0.063102
MS-Variance AR(1) + geopolitical risk	0.262384	0.181056	0.066281
MS-Variance AR(1) + energy risk	0.269213	0.182300	0.081396
MS-Variance AR(1) + supply-chain risk	0.263819	0.181574	0.070967
MS-Variance AR(1) + all subject risks	0.274330	0.185814	0.091666
Static AR-HMM	0.262183	0.181303	0.074565
AR-HMM(1) + geopolitical risk	0.271333	0.190991	0.091366
AR-HMM(1) + energy risk	0.278929	0.183632	0.101043
AR-HMM(1) + supply-chain risk	0.264284	0.183961	0.078634
AR-HMM(1) + all subject risks	0.284345	0.197615	0.104262

Topic risks use lag 1. This separate saved bundle contains a different unrestricted AR-HMM fit from Table A.5; comparisons should use each bundle's own baseline.